Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.41_3)

Metric

$$ds^2 = -H(v,z)^2\,\mathrm{d}u\otimes\mathrm{d}u + \left(-\frac{l^2-z^2}{l^2+z^2}\right)\mathrm{d}u\otimes\mathrm{d}v + \left(-\frac{l^2vz+2\,mvz^2-vz^3}{l^4+2\,l^2z^2+z^4}\right)\mathrm{d}u\otimes\mathrm{d}z + \left(\frac{l^3v+2\,lmvz-lvz^2}{l^4+2\,l^2z^2+z^4}\right)\mathrm{d}u\otimes\mathrm{d}y + \left(-\frac{l^2-z^2}{l^2+z^2}\right)\mathrm{d}v \otimes\mathrm{d}u + \left(-\frac{l^2vz+2\,mvz^2-vz^3}{l^4+2\,l^2z^2+z^4}\right)\mathrm{d}z\otimes\mathrm{d}u + \mathrm{d}z\otimes\mathrm{d}z + \left(\frac{l^3v+2\,lmvz-lvz^2}{l^4+2\,l^2z^2+z^4}\right)\mathrm{d}y\otimes\mathrm{d}u + \left(\frac{l^2+z^2}{l^2+2\,mz-z^2}\right)\mathrm{d}y\otimes\mathrm{d}y$$

Coordinates

$(x^\mu) = (u, v, z, y)$

Metric Functions

$$H = -\frac{2\,l^4v^2}{l^6+3\,l^4z^2+3\,l^2z^4+z^6} - \frac{4\,l^2mv^2z}{l^6+3\,l^4z^2+3\,l^2z^4+z^6} + \frac{2\,l^2v^2z^2}{l^6+3\,l^4z^2+3\,l^2z^4+z^6} + \frac{v^2}{2\,(l^2+z^2)}$$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(\frac{1}{2}\,H(v,z)^2\right)\,du + \left(\frac{l^2}{l^2+z^2} - \frac{z^2}{l^2+z^2}\right)\,dv + \left(\frac{l^2vz}{l^4+2\,l^2z^2+z^4} + \frac{2\,mvz^2}{l^4+2\,l^2z^2+z^4} - \frac{vz^3}{l^4+2\,l^2z^2+z^4}\right)\,dz + \left(-\frac{l^3v}{l^4+2\,l^2z^2+z^4} - \frac{2\,lmvz}{l^4+2\,l^2z^2+z^4} + \frac{lvz^2}{l^4+2\,l^2z^2+z^4}\right)\,dy$$

$$m = \left(\sqrt{\frac{1}{2}}\right)\,dz + \left(\frac{i\,\sqrt{\frac{1}{2}}\sqrt{l^2+z^2}}{\sqrt{l^2+2\,mz-z^2}}\right)\,dy$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{\frac{1}{2}}\sqrt{l^2+z^2}}{\sqrt{l^2+2\,mz-z^2}}\right) dy$$

Contravariant components

$$l = \left(-\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\right) dv$$

$$n = (-1)\, du + \left(\frac{l^2 H(v,z)^2}{2\,(l^2-z^2)} + \frac{z^2 H(v,z)^2}{2\,(l^2-z^2)}\right) dv$$

$$\begin{aligned} m = \Bigg(& \frac{i\sqrt{2}\sqrt{l^2+z^2}l^5 v}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}l^4 vz}{2\,(l^6+l^4z^2-l^2z^4-z^6)} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3 mvz}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3 vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} \\ & - \frac{\sqrt{2}l^2 mvz^2}{l^6+l^4z^2-l^2z^4-z^6} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2 vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} \\ & - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} + \frac{\sqrt{2}vz^5}{2\,(l^6+l^4z^2-l^2z^4-z^6)}\Bigg) dv + \left(\sqrt{\frac{1}{2}}\right) dz + \left(\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}}{2\sqrt{l^2+z^2}}\right) dy \end{aligned}$$

$$\begin{aligned} \overline{m} = \Bigg(& -\frac{i\sqrt{2}\sqrt{l^2+z^2}l^5 v}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}l^4 vz}{2\,(l^6+l^4z^2-l^2z^4-z^6)} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3 mvz}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3 vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} \\ & - \frac{\sqrt{2}l^2 mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2 vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{l^2+2\,mz-z^2}} \\ & - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} + \frac{\sqrt{2}vz^5}{2\,(l^6+l^4z^2-l^2z^4-z^6)}\Bigg) dv + \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}}{2\sqrt{l^2+z^2}}\right) dy \end{aligned}$$

Directional Derivatives

$$D = -\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\,\partial_v$$

$$\Delta = -\partial_u + \frac{l^2 H(v,z)^2}{2\left(l^2 - z^2\right)} + \frac{z^2 H(v,z)^2}{2\left(l^2 - z^2\right)}\,\partial_v$$

$$\begin{aligned}\delta = {} & \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^5 v}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}\,l^4 vz}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,l^3 mvz}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^3 vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} \\ & - \frac{\sqrt{2}\,l^2 mvz^2}{l^6+l^4z^2-l^2z^4-z^6} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,lm^2vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,lmvz^3}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} \\ & + \frac{i\sqrt{2}\sqrt{l^2+z^2}\,lvz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}\,mvz^4}{l^6+l^4z^2-l^2z^4-z^6} + \frac{\sqrt{2}\,vz^5}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}\,\partial_v + \sqrt{\frac{1}{2}}\,\partial_z + \frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}}{2\sqrt{l^2+z^2}}\,\partial_y\end{aligned}$$

$$\begin{aligned}\overline{\delta} = {} & -\frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^5 v}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}\,l^4 vz}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,l^3 mvz}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} \\ & + \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^3 vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}\,l^2 mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,lm^2vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,lmvz^3}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} \\ & - \frac{i\sqrt{2}\sqrt{l^2+z^2}\,lvz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2\,mz-z^2}} - \frac{\sqrt{2}\,mvz^4}{l^6+l^4z^2-l^2z^4-z^6} + \frac{\sqrt{2}\,vz^5}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}\,\partial_v + \sqrt{\frac{1}{2}}\,\partial_z - \frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}}{2\sqrt{l^2+z^2}}\,\partial_y\end{aligned}$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned}\mu = {} & \frac{i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,l^5 mv}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} - \frac{i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,l^5 vz}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} + \frac{2i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,l^3 vz^3}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} \\ & + \frac{3i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,lmvz^4}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} - \frac{i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,lvz^5}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}}\end{aligned}$$

$$\lambda = 0$$

$$
\begin{aligned}
\nu = & -\frac{i\sqrt{2}\sqrt{l^2+z^2}l^5 vH(v,z)\frac{\partial}{\partial v}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} - \frac{\sqrt{2}l^4 vzH(v,z)\frac{\partial}{\partial v}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3 mvzH(v,z)\frac{\partial}{\partial v}H(v,z)}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3 vz^2H(v,z)\frac{\partial}{\partial v}H(v,z)}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} \\
& - \frac{\sqrt{2}l^2 mvz^2H(v,z)\frac{\partial}{\partial v}H(v,z)}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2H(v,z)\frac{\partial}{\partial v}H(v,z)}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3H(v,z)\frac{\partial}{\partial v}H(v,z)}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4H(v,z)\frac{\partial}{\partial v}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} \\
& - \frac{\sqrt{2}mvz^4H(v,z)\frac{\partial}{\partial v}H(v,z)}{l^6+l^4z^2-l^2z^4-z^6} + \frac{\sqrt{2}vz^5H(v,z)\frac{\partial}{\partial v}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{\sqrt{2}l^6H(v,z)\frac{\partial}{\partial z}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{\sqrt{2}l^4z^2H(v,z)\frac{\partial}{\partial z}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{\sqrt{2}l^2z^4H(v,z)\frac{\partial}{\partial z}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} \\
& - \frac{\sqrt{2}z^6H(v,z)\frac{\partial}{\partial z}H(v,z)}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^5H(v,z)^2}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{5\sqrt{2}l^4zH(v,z)^2}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3mzH(v,z)^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} \\
& - \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3z^2H(v,z)^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{\sqrt{2}l^2mz^2H(v,z)^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{i\sqrt{2}\sqrt{l^2+z^2}lm^2z^2H(v,z)^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{\sqrt{2}l^2z^3H(v,z)^2}{l^6+l^4z^2-l^2z^4-z^6} \\
& - \frac{i\sqrt{2}\sqrt{l^2+z^2}lmz^3H(v,z)^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}lz^4H(v,z)^2}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{l^2+2mz-z^2}} + \frac{\sqrt{2}mz^4H(v,z)^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{\sqrt{2}z^5H(v,z)^2}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}
\end{aligned}
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\begin{aligned}
\alpha = & \frac{l^6m}{2\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} + \frac{l^6z}{4\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& + \frac{5\,l^4mz^2}{2\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} - \frac{l^4z^3}{4\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& + \frac{l^2m^2z^3}{\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8} + \frac{3\,l^2mz^4}{2\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& - \frac{l^2z^5}{4\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} + \frac{m^2z^5}{\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8} \\
& - \frac{mz^6}{2\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} + \frac{z^7}{4\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& + \frac{i\sqrt{l^2+2mz-z^2}\sqrt{l^2+z^2}l^5}{4\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} + \frac{i\sqrt{l^2+2mz-z^2}\sqrt{l^2+z^2}l^3mz}{\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8} \\
& - \frac{i\sqrt{l^2+2mz-z^2}\sqrt{l^2+z^2}l^3z^2}{2\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)} + \frac{i\sqrt{l^2+2mz-z^2}\sqrt{l^2+z^2}lm^2z^2}{\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8} \\
& - \frac{i\sqrt{l^2+2mz-z^2}\sqrt{l^2+z^2}lmz^3}{\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8} + \frac{i\sqrt{l^2+2mz-z^2}\sqrt{l^2+z^2}lz^4}{4\left(\sqrt{2}l^8+2\sqrt{2}l^6mz+2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4-2\sqrt{2}l^2mz^5-2\sqrt{2}mz^7+\sqrt{2}z^8\right)}
\end{aligned}
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\begin{aligned}
\beta = &-\frac{i\sqrt{l^2+2\,mz-z^2}\,l^5}{4\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}}-\frac{l^4m}{2\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)}\\
&+\frac{9\,l^4z}{4\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)}-\frac{i\sqrt{l^2+2\,mz-z^2}\,l^3mz}{\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}}\\
&+\frac{i\sqrt{l^2+2\,mz-z^2}\,l^3z^2}{2\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}}\\
&+\frac{4\,l^2mz^2}{\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6}-\frac{i\sqrt{l^2+2\,mz-z^2}\,lm^2z^2}{\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}}\\
&-\frac{5\,l^2z^3}{2\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)}+\frac{i\sqrt{l^2+2\,mz-z^2}\,lmz^3}{\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}}\\
&+\frac{m^2z^3}{\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6}-\frac{i\sqrt{l^2+2\,mz-z^2}\,lz^4}{4\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}}\\
&-\frac{3\,mz^4}{2\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)}+\frac{z^5}{4\left(\sqrt{2}l^6+2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-2\sqrt{2}mz^5+\sqrt{2}z^6\right)}
\end{aligned}
$$

$$
\tau = -\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}\,l^3}{4\left(l^4-z^4\right)\sqrt{l^2+z^2}}+\frac{5\sqrt{2}l^2z}{4\left(l^4-z^4\right)}-\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}\,lmz}{2\left(l^4-z^4\right)\sqrt{l^2+z^2}}+\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}\,lz^2}{4\left(l^4-z^4\right)\sqrt{l^2+z^2}}+\frac{\sqrt{2}mz^2}{2\left(l^4-z^4\right)}-\frac{\sqrt{2}z^3}{4\left(l^4-z^4\right)}
$$

$$
\begin{aligned}
\pi = &-\frac{5\sqrt{2}l^4z}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}-\frac{\sqrt{2}l^2mz^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}-\frac{\sqrt{2}l^2z^3}{l^6+l^4z^2-l^2z^4-z^6}-\frac{\sqrt{2}mz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}+\frac{\sqrt{2}z^5}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}\\
&-\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,l^3}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}-\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,lmz}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}+\frac{i\sqrt{2}\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}\,lz^2}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}
\end{aligned}
$$

$$
\epsilon = 0
$$

$$
\begin{aligned}
\gamma = & -\frac{l^8 H(v,z)\frac{\partial}{\partial v}H(v,z)}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)} - \frac{2\,l^6z^2H(v,z)\frac{\partial}{\partial v}H(v,z)}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{3\,l^4z^4H(v,z)\frac{\partial}{\partial v}H(v,z)}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{2\,l^2z^6H(v,z)\frac{\partial}{\partial v}H(v,z)}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} \\
& - \frac{z^8H(v,z)\frac{\partial}{\partial v}H(v,z)}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)} + \frac{i\,l^7mv}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} - \frac{i\,l^7vz}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{i\,l^5m^2vz}{\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} - \frac{3i\,l^5mvz^2}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} + \frac{3i\,l^5vz^3}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{7i\,l^3mvz^4}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} - \frac{3i\,l^3vz^5}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} + \frac{3i\,lm^2vz^5}{\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& - \frac{5i\,lmvz^6}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} + \frac{i\,lvz^7}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}
\end{aligned}
$$

Ricci Tensor Components

$\Phi_{00} = 0$

$\Phi_{01} = 0$

$$
\begin{aligned}
\Phi_{02} = & \frac{11\,l^{10}}{8\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} + \frac{23\,l^8mz}{4\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& - \frac{55\,l^8z^2}{8\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} + \frac{11\,l^6m^2z^2}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& - \frac{53\,l^6mz^3}{4\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} + \frac{4\,l^4m^3z^3}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& + \frac{5\,l^6z^4}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} - \frac{10\,l^4m^2z^4}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& + \frac{2\,l^2m^4z^4}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} + \frac{19\,l^4mz^5}{4\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& - \frac{5\,l^2m^3z^5}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} + \frac{11\,l^4z^6}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& + \frac{l^2m^2z^6}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} + \frac{53\,l^2mz^7}{4\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& - \frac{m^3z^7}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} - \frac{43\,l^2z^8}{8\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& + \frac{4\,m^2z^8}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} - \frac{5\,mz^9}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& + \frac{3\,z^{10}}{8\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} - \frac{i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}l^7m}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& + \frac{11i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}l^7z}{4\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} - \frac{2i\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}l^5m^2z}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} + \ldots
\end{aligned}
$$

$\Phi_{10} = 0$

$$
\begin{aligned}
\Phi_{11} = & \frac{l^{14} \frac{\partial}{\partial v} H(v,z)^2}{4\,(l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14})} \\
& + \frac{l^{12}mz \frac{\partial}{\partial v} H(v,z)^2}{l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14}} \\
& + \frac{3\,l^{12}z^2 \frac{\partial}{\partial v} H(v,z)^2}{4\,(l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14})} \\
& + \frac{l^{10}m^2z^2 \frac{\partial}{\partial v} H(v,z)^2}{l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14}} \\
& + \frac{4\,l^{10}mz^3 \frac{\partial}{\partial v} H(v,z)^2}{l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14}} \\
& + \frac{l^{10}z^4 \frac{\partial}{\partial v} H(v,z)^2}{4\,(l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14})} \\
& + \frac{5\,l^8m^2z^4 \frac{\partial}{\partial v} H(v,z)^2}{l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14}} \\
& + \frac{5\,l^8mz^5 \frac{\partial}{\partial v} H(v,z)^2}{l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14}} \\
& - \frac{5\,l^8z^6 \frac{\partial}{\partial v} H(v,z)^2}{4\,(l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14})} \\
& + \frac{10\,l^6m^2z^6 \frac{\partial}{\partial v} H(v,z)^2}{l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14}} \\
& - \frac{5\,l^6z^8 \frac{\partial}{\partial v} H(v,z)^2}{4\,(l^{14} + 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 - 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 + 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} - 4\,mz^{13} + z^{14})} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = &-\frac{i\sqrt{2}l^{15}v\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{i\sqrt{2}l^{13}mvz\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{3i\sqrt{2}l^{13}vz^2\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{i\sqrt{2}l^{11}m^2vz^2\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{4i\sqrt{2}l^{11}mvz^3\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{i\sqrt{2}l^{11}vz^4\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{5i\sqrt{2}l^9m^2vz^4\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{5i\sqrt{2}l^9mvz^5\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&+\frac{5i\sqrt{2}l^9vz^6\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{10i\sqrt{2}l^7m^2vz^6\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&+\frac{5i\sqrt{2}l^7vz^8\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{10i\sqrt{2}l^5m^2vz^8\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&+\frac{5i\sqrt{2}l^5mvz^9\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{i\sqrt{2}l^5vz^{10}\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{5i\sqrt{2}l^3m^2vz^{10}\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&+\frac{4i\sqrt{2}l^3mvz^{11}\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}\\
&-\frac{3i\sqrt{2}l^3vz^{12}\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}}-\cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{11\,l^{10}}{8\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} + \frac{23\,l^{8}mz}{4\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& - \frac{55\,l^{8}z^{2}}{8\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} + \frac{11\,l^{6}m^{2}z^{2}}{2\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& - \frac{53\,l^{6}mz^{3}}{4\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} + \frac{4\,l^{4}m^{3}z^{3}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} \\
& + \frac{5\,l^{6}z^{4}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} - \frac{10\,l^{4}m^{2}z^{4}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} \\
& + \frac{2\,l^{2}m^{4}z^{4}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} + \frac{19\,l^{4}mz^{5}}{4\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& - \frac{5\,l^{2}m^{3}z^{5}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} + \frac{11\,l^{4}z^{6}}{2\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& + \frac{l^{2}m^{2}z^{6}}{2\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} + \frac{53\,l^{2}mz^{7}}{4\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& - \frac{m^{3}z^{7}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} - \frac{43\,l^{2}z^{8}}{8\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& + \frac{4\,m^{2}z^{8}}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} - \frac{5\,mz^{9}}{2\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} \\
& + \frac{3\,z^{10}}{8\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} + \frac{i\,\sqrt{l^{2}+2\,mz-z^{2}}\sqrt{l^{2}+z^{2}}l^{7}m}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} \\
& - \frac{11i\,\sqrt{l^{2}+2\,mz-z^{2}}\sqrt{l^{2}+z^{2}}l^{7}z}{4\,(l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12})} + \frac{2i\,\sqrt{l^{2}+2\,mz-z^{2}}\sqrt{l^{2}+z^{2}}l^{5}m^{2}z}{l^{12}+2\,l^{10}mz+2\,l^{8}mz^{3}-3\,l^{8}z^{4}-4\,l^{6}mz^{5}-4\,l^{4}mz^{7}+3\,l^{4}z^{8}+2\,l^{2}mz^{9}+2\,mz^{11}-z^{12}} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = & \frac{i\sqrt{2}l^{15}v\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}l^{13}mvz\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{3i\sqrt{2}l^{13}vz^2\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}l^{11}m^2vz^2\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{4i\sqrt{2}l^{11}mvz^3\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}l^{11}vz^4\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{9}m^2vz^4\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{9}mvz^5\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& - \frac{5i\sqrt{2}l^{9}vz^6\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{10i\sqrt{2}l^{7}m^2vz^6\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& - \frac{5i\sqrt{2}l^{7}vz^8\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{10i\sqrt{2}l^{5}m^2vz^8\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& - \frac{5i\sqrt{2}l^{5}mvz^9\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}l^{5}vz^{10}\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{3}m^2vz^{10}\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& - \frac{4i\sqrt{2}l^{3}mvz^{11}\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} \\
& + \frac{3i\sqrt{2}l^{3}vz^{12}\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}\sqrt{l^2+z^2}} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{l^{18} v^2 \frac{\partial}{\partial v} H(v,z)^2}{2\,(l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20})} \\
& + \frac{4\,l^{16} mv^2 z \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& + \frac{l^{16} v^2 z^2 \frac{\partial}{\partial v} H(v,z)^2}{2\,(l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20})} \\
& + \frac{12\,l^{14} m^2 v^2 z^2 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& + \frac{7\,l^{14} mv^2 z^3 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& + \frac{16\,l^{12} m^3 v^2 z^3 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& - \frac{l^{14} v^2 z^4 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& + \frac{30\,l^{12} m^2 v^2 z^4 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& + \frac{8\,l^{10} m^4 v^2 z^4 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& - \frac{3\,l^{12} mv^2 z^5 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} \\
& + \frac{52\,l^{10} m^3 v^2 z^5 \frac{\partial}{\partial v} H(v,z)^2}{l^{20} + 2\,l^{18} mz + 4\,l^{18} z^2 + 10\,l^{16} mz^3 + 3\,l^{16} z^4 + 16\,l^{14} mz^5 - 8\,l^{14} z^6 - 14\,l^{12} z^8 - 28\,l^{10} mz^9 - 28\,l^8 mz^{11} + 14\,l^8 z^{12} + 8\,l^6 z^{14} + 16\,l^4 mz^{15} - 3\,l^4 z^{16} + 10\,l^2 mz^{17} - 4\,l^2 z^{18} + 2\,mz^{19} - z^{20}} - \dots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & -\frac{l^{14} \frac{\partial}{\partial v} H(v,z)^2}{12\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{l^{12} mz \frac{\partial}{\partial v} H(v,z)^2}{3\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{l^{12} z^2 \frac{\partial}{\partial v} H(v,z)^2}{4\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{l^{10} m^2 z^2 \frac{\partial}{\partial v} H(v,z)^2}{3\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{4\,l^{10} mz^3 \frac{\partial}{\partial v} H(v,z)^2}{3\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{l^{10} z^4 \frac{\partial}{\partial v} H(v,z)^2}{12\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{5\,l^8 m^2 z^4 \frac{\partial}{\partial v} H(v,z)^2}{3\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{5\,l^8 mz^5 \frac{\partial}{\partial v} H(v,z)^2}{3\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& + \frac{5\,l^8 z^6 \frac{\partial}{\partial v} H(v,z)^2}{12\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& - \frac{10\,l^6 m^2 z^6 \frac{\partial}{\partial v} H(v,z)^2}{3\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} \\
& + \frac{5\,l^6 z^8 \frac{\partial}{\partial v} H(v,z)^2}{12\,(l^{14} + 4\,l^{12} mz - l^{12} z^2 + 4\,l^{10} m^2 z^2 - 3\,l^{10} z^4 + 4\,l^8 m^2 z^4 - 12\,l^8 mz^5 + 3\,l^8 z^6 - 8\,l^6 m^2 z^6 + 3\,l^6 z^8 - 8\,l^4 m^2 z^8 + 12\,l^4 mz^9 - 3\,l^4 z^{10} + 4\,l^2 m^2 z^{10} - l^2 z^{12} + 4\,m^2 z^{12} - 4\,mz^{13} + z^{14})} - \dots
\end{aligned}
$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = {} & \frac{l^{14}\frac{\partial}{\partial v}H(v,z)^2}{6\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{2\,l^{12}mz\frac{\partial}{\partial v}H(v,z)^2}{3\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{l^{12}z^2\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{2\,l^{10}m^2z^2\frac{\partial}{\partial v}H(v,z)^2}{3\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{8\,l^{10}mz^3\frac{\partial}{\partial v}H(v,z)^2}{3\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{l^{10}z^4\frac{\partial}{\partial v}H(v,z)^2}{6\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{10\,l^8m^2z^4\frac{\partial}{\partial v}H(v,z)^2}{3\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{10\,l^8mz^5\frac{\partial}{\partial v}H(v,z)^2}{3\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& - \frac{5\,l^8z^6\frac{\partial}{\partial v}H(v,z)^2}{6\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& + \frac{20\,l^6m^2z^6\frac{\partial}{\partial v}H(v,z)^2}{3\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} \\
& - \frac{5\,l^6z^8\frac{\partial}{\partial v}H(v,z)^2}{6\left(l^{14}+4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^8m^2z^4-12\,l^8mz^5+3\,l^8z^6-8\,l^6m^2z^6+3\,l^6z^8-8\,l^4m^2z^8+12\,l^4mz^9-3\,l^4z^{10}+4\,l^2m^2z^{10}-l^2z^{12}+4\,m^2z^{12}-4\,mz^{13}+z^{14}\right)} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = {} & \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^{13}v\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} + \frac{\sqrt{2}l^{12}vz\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)} \\
& + \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^{11}mvz\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^{11}vz^2\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& + \frac{\sqrt{2}l^{10}mvz^2\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)} + \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^9m^2vz^2\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& + \frac{\sqrt{2}l^{10}vz^3\frac{\partial}{\partial v}H(v,z)^2}{l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}} + \frac{3i\sqrt{2}\sqrt{l^2+z^2}\,l^9mvz^3\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& - \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^9vz^4\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& + \frac{5\sqrt{2}l^8mvz^4\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)} + \frac{4i\sqrt{2}\sqrt{l^2+z^2}\,l^7m^2vz^4\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& + \frac{5\sqrt{2}l^8vz^5\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,l^7mvz^5\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& - \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^7vz^6\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} + \frac{5\sqrt{2}l^6mvz^6\frac{\partial}{\partial v}H(v,z)^2}{l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}} \\
& + \frac{6i\sqrt{2}\sqrt{l^2+z^2}\,l^5m^2vz^6\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}\,l^5mvz^7\frac{\partial}{\partial v}H(v,z)^2}{\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} \\
& - \frac{i\sqrt{2}\sqrt{l^2+z^2}\,l^5vz^8\frac{\partial}{\partial v}H(v,z)^2}{4\left(l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}\right)\sqrt{l^2+2\,mz-z^2}} + \frac{5\sqrt{2}l^4mvz^8\frac{\partial}{\partial v}H(v,z)^2}{l^{14}+3\,l^{12}z^2+l^{10}z^4-5\,l^8z^6-5\,l^6z^8+l^4z^{10}+3\,l^2z^{12}+z^{14}} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = & -\frac{l^{10}v^2\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& -\frac{4\,l^8mv^2z\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{5\,l^8v^2z^2\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& -\frac{12\,l^6m^2v^2z^2\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{15\,l^6mv^2z^3\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& -\frac{16\,l^4m^3v^2z^3\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& -\frac{4\,l^6v^2z^4\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{30\,l^4m^2v^2z^4\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& -\frac{8\,l^2m^4v^2z^4\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& -\frac{15\,l^4mv^2z^5\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{20\,l^2m^3v^2z^5\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{2\,l^4v^2z^6\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& -\frac{12\,l^2m^2v^2z^6\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{l^2mv^2z^7\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{4\,m^3v^2z^7\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{l^2v^2z^8\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} \\
& -\frac{6\,m^2v^2z^8\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& +\frac{3\,mv^2z^9\frac{\partial}{\partial v}H(v,z)^2}{l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}} \\
& -\frac{v^2z^{10}\frac{\partial}{\partial v}H(v,z)^2}{2\left(l^{12}+2\,l^{10}mz+2\,l^8mz^3-3\,l^8z^4-4\,l^6mz^5-4\,l^4mz^7+3\,l^4z^8+2\,l^2mz^9+2\,mz^{11}-z^{12}\right)} - \cdots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"